Million-Token Agentic-RL Infrastructure

Deep Dive into Kimi K3 — Chapter 42

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Retire accelerator state freely — but keep the transcript
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Goal: follow one paused repair attempt, G4, through co-located phases, prefix retention, admission, buffer reuse, microVM snapshots, and generation-fenced recovery under a new policy version. [REPORTED: K3 Sec. 5.3; fixtures synthetic]

Roadmap

Three states at the pause

Memory across phases

Sandboxes and recovery

Trace and accounting

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Transcript, sandbox, cache: three owners

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durable; v7
scores kept

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sandbox E8
patched file; restore

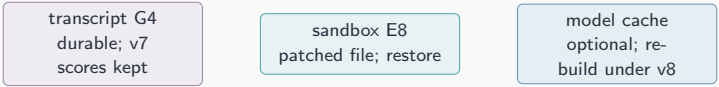
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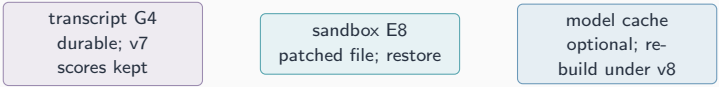
model cache
optional; re-
build under v8

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- Useful completion counts admitted complete groups — recomputed tokens cost resources but are not new samples.

Memory across phases

Occupancy, reservations, and lifetimes

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- Two gradient slots: 11 ms ideal versus 15 ms serial; a late reader pushes completion to 12 ms. [DERIVED: ideal fixture, not measured]

Sandboxes and recovery

Ordered snapshots and fenced generations

- E8 = base $\rightarrow D_1 \rightarrow D_2$ plus CPU, device, disk, and configuration; page 3 comes from D_2; a fork's three writes add 12 KiB. [REPORTED: K3 Sec. 5.3.2 AgentENV]

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- Recovery: prepare, validate, commit, admit; stale generation 5 is fenced, generation 6 resumes; unknown external effects stay blocked.

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One trace, honest accounting

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Next: Chapter 43, Mooncake and Disaggregated Serving.